

Lab4 M1 & Syscalls

BYU Electrical & Computer
Engineering
IRA A. FULTON COLLEGE OF ENGINEERING

- Implement Space Invaders
- You will work in a group
 - Groups will be assigned today.
 - You may swap members of a group (if both sides agree) until end of day Friday.
 - Message me on teams.
- Team repositories will be created this weekend

Milestone 1

Implement graphics

Three test programs:

1. graphics_test
2. graphics_syscalls
(Less than 500 syscalls)
3. graphics_benchmarking
(must draw sprites in <3ms)



What is a System Call?

What happens when you call `read(fd, buf, 100)`?

- Your program calls `read()`. That's a C library function, still in user mode.
- The wrapper puts the system call number in a register and the arguments in others, then executes **`svc`** (ARM trap instruction)
- The CPU switches to kernel mode and jumps to the exception vector. Every register is saved.
- The kernel looks up the handler, validates the file descriptor, and checks that your buffer address is really yours.
- The filesystem finds the data, already in memory from the last read, and copies 100 bytes into your buffer.
- Registers are restored, the CPU drops back to user mode, and `read()` returns.

What changes for a read of 1 byte or 100,000 bytes?

Demo 1

See examples/syscall/ in student repo

open vs fopen, read vs fread

- `fopen()` calls `open()` for you, then allocates a `FILE` struct: the file descriptor, a 4 KB buffer, and a pointer to where you are in that buffer.
- `fread()` copies from that buffer. Only when the buffer runs dry does it make one `read()` call for the next 4 KB.

`examples/syscall/fread.c`

<https://byu-cpe.github.io/ecen427/documentation/hdmi/>

Useful syscalls:

- lseek
- write
- read
- open/close

Demo 2

Instructor demo of `userspace/apps/frame_test`

Lab 4 M1 Apps & Classes

- https://github.com/byu-cpe/ecen427_student/blob/main/userspace/apps/graphics_test/main.cpp
- https://github.com/byu-cpe/ecen427_student/blob/main/userspace/apps/space_invaders/Graphics.h
- https://github.com/byu-cpe/ecen427_student/blob/main/userspace/apps/space_invaders/Sprites.h
- https://github.com/byu-cpe/ecen427_student/blob/main/userspace/apps/space_invaders/Sprites.cpp
- https://github.com/byu-cpe/ecen427_student/blob/main/userspace/apps/space_invaders/resources/sprites.c